Power Budget Example

Team Number:	307
Project Name:	**3D POSTIONMENT**
Team Member Names:	**Garret, Jt, Abriana,Lis,Zane**
Version:	**Donovan Calderon**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	ESP32-S3-WROOM-1-N4	ESP32-S3-WROOM-1-N4	+3 - 3.6V	1	400	400	mA
	AS5600L	AS5600L-ASOM SOIC8	+3 - 3.3V	2	6	12	mA
	AS5600	AS5600-ASOM SOIC8	+3 - 3.3V	2	6	12	mA
	Servo Motor	MG995	+5 - 6V	2	2500	5000	mA
	LEDs	LTST-C150KT	-5V - +2.1V	4	20	80	mA
					0	0	mA
+5V Power Rail	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	Servo Motors	MG995	5 - 6V	2	2000	4000	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	4000	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	5000	mA
c2. Regulator or Source Ch	+5V Regulator	TPS56524DRLR	5V	1	5000	5000	mA
					Total Remaining Current Available on +5V Rail	0	mA
+3.3V Power Rail	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**Total**	**Unit**
	Esp32-S3	ESP32-S3-WROOM-1-N4	+1.8 - 3.3V	1	375	375	mA
	AS5600	AS5600-ASOM SOIC8	+3 - 3.3V	2	6	12	mA
	AS5600L	AS5600L-ASOM SOIC8	+3 - 3.3V	2	6	12	mA
	LEDs	LTST-C150KT	+2 - 2.2V	4	10	40	mA
					Subtotal	439	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	548.75	mA
c4. Regulator or Source Ch	+3.3V low-dropout regulat	LM2575T-3.3G	+5V - 20V	1	1000	1000	mA
					Total Remaining Current Available on 3.3V Rail	451.25	mA

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If

External Power Source 1	**Component Name**	**Part Number**	**SupplyVoltageRange**	**Outpu**	**AbsoluteMaximumCurrent (mA)**	**Total**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	GSD-120500	240VAC	12	5000	60000	mA
						0	mA
Power Rails Connected to External Power Source 1	+5V Regulator	TPS56524DRLR	+3 - 16V	5	5000	25000	mA
	+3.3V low-dropout regulat	LM2575T-3.3G	+4.75V - 40V	3.3	1000	3300	mA
					Total Remaining Current Available on External Power Source 1	31700	mA